

INVITED TALK / ABSTRACT

PIC1 pinched manifolds are flat or compact

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Abstract

Hamilton's pinching conjecture, that three-dimensional complete non-compact manifolds with pinched Ricci curvature are flat, was resolved recently using Ricci flow. In this talk I will explain a direct analogue of that result in all dimensions. One aspect of the work is a new lifting technique that allows us to handle manifolds that are collapsed at infinity; until now this could only be achieved in 3D via work of Lott. The new result builds on earlier work of Deruelle, Simon and the speaker and separately of Lee-Topping. Joint work with Alix Deruelle, Man Chun Lee, Miles Simon and Peter Topping.

Keywords: PIC1 pinched manifolds; Hamilton's pinching conjecture; pinched Ricci curvature; Ricci flow; non-compact manifolds; lifting technique; collapsed at infinity